CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION

Cambridge IGCSE™ & O Level Computer Science (0478 / 2210)

TOPIC 1.2: TEXT, SOUND AND IMAGES • EXAM PATTERN WORKBOOK

Student Name: _____ Class: O Level / Grade 10 Total Marks: 50 Marks

PART A: TEXT REPRESENTATION & CHARACTER SETS

Question 1 Cambridge Style Past Paper [5 marks]

(a) Define what is meant by a **character set**. [2]

Definition:

(b) Describe two differences between the **ASCII** character set and the **Unicode** character set. [2]

Differences:

(c) State one reason why Unicode is now preferred over standard ASCII in modern computer systems. [1]

Reason:

Question 2 Calculation Problem [4 marks]

A short story document contains **4,096 characters**. Each character is encoded using 8-bit Extended ASCII.

(a) Calculate the file size of the text document in **Bytes**. [1]

Working:

(b) Calculate the file size in **Kibibytes (KiB)**. Show your full working. [3]

Working & Final Answer:

PART B: SOUND REPRESENTATION & SAMPLING

Question 3 Theory & Concept [6 marks]

(a) Explain how continuous analogue sound waves are converted into digital sound files for computer storage. [3]

Explanation:

(b) State the hardware device used to convert analogue sound signals into digital data. [1]

Device:

(c) Define the terms: [2]

• **Sample Rate:** _____

• **Sample Resolution (Bit Depth):** _____

Question 4 Calculation Problem [5 marks]

A music producer records a **20-second stereo audio track** (2 channels). The recording settings are:

• Sample Rate = **44,100 Hz** • Sample Resolution = **16 bits**

(a) Calculate the total file size in **bits**. [2]

Working:

(b) Calculate the uncompressed file size in **Megabytes (MB)**. Show your working. [3]

Working & Final Answer:

PART C: IMAGE REPRESENTATION & BITMAPS

Question 5 Theory & Definitions [6 marks]

(a) Define the following image terms: [3]

• **Pixel:** _____

• **Image Resolution:** _____

• **Colour Depth:** _____

(b) State the number of unique colours that can be represented using a colour depth of: [2]

(i) 4 bits: _____ (ii) 8 bits: _____

(c) Give one example of **metadata** stored in an image file. [1]

Example:

Question 6 Calculation Problem [5 marks]

A digital camera captures an uncompressed bitmap photograph with a resolution of **2000 × 1500 pixels** and a colour depth of **24 bits**.

(a) Calculate the total image size in **Bytes**. [2]

Working:

(b) Calculate the image size in **Mebibytes (MiB)**. Show your working. [3]

Working & Final Answer:

PART D: DATA COMPRESSION (LOSSY, LOSSLESS & RLE)

Question 7 Comparison & Analysis [6 marks]

(a) Complete the table below by ticking (✓) one box for each file type to show whether lossy or lossless compression is most suitable. [3]

File Description	Lossy Compression	Lossless Compression
A high-resolution photograph to upload on a social media website (JPEG)	[ ]	[ ]
A computer program executable file (program.exe)	[ ]	[ ]
A music track for an online music streaming app (MP3)	[ ]	[ ]

(b) Explain why lossless compression must be used for text files and computer programs. [2]

Explanation:

(c) State one benefit of file compression for streaming video over the Internet. [1]

Benefit:

Question 8 Run-Length Encoding (RLE) [5 marks]

(a) Describe how Run-Length Encoding (RLE) compresses data. [2]

Description:

(b) The following row of 16 pixels from a black-and-white icon is given:

```
W W W W W W   B B B   W W W W   B B B
```

Show how this pixel row is represented using Run-Length Encoding. [3]

RLE Representation:

PART E: MULTIPLE CHOICE EXAM DRILL

Question 9 Quick MCQ Drill [8 marks]

1. How many bits are used in Standard ASCII? [ ] a) 7 bits [ ] b) 8 bits [ ] c) 16 bits
2. What is the unit of measurement for Sample Rate? [ ] a) Pixels [ ] b) Hertz (Hz) [ ] c) Bits
3. An image with 8-bit colour depth can display how many colours? [ ] a) 16 [ ] b) 256 [ ] c) 65,536
4. Which of the following is an example of Lossless compression? [ ] a) MP3 [ ] b) JPEG [ ] c) PNG / RLE
